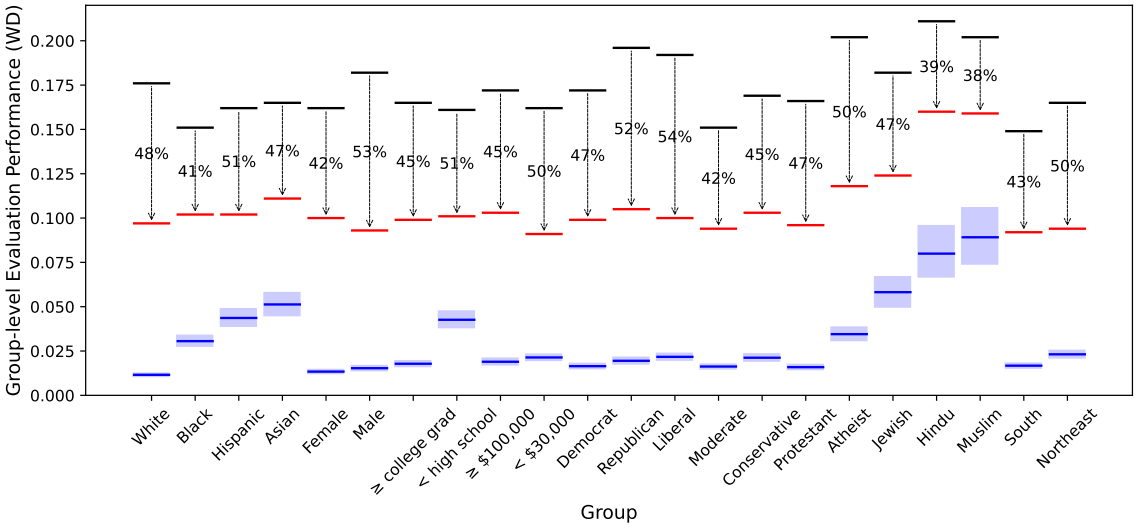

Figure 3: Per-group evaluation performance of our model Llama-2-7B-SubPOP-FT (red lines) on OpinionQA. For comparison, the results from zero-shot QA prompting (black lines) and the lower bound (blue lines) are presented. We observe that the relative improvement, measuring how much of the gap between zero-shot prompting and the lower bound has been closed, remains consistent across subpopulations. Shaded blue regions represent the 95% confidence interval of the lower-bound estimation for each group. Per-group results for other models (Table 9) and the results on SubPOP evaluation set (Table 10) are available in Appendix E.

4.3 Generalization across Target Subpopulations

Here we report two key observations: (1) prediction performance improves consistently across most subpopulations represented in the fine-tuning data, and (2) the LLMs fine-tuned on SubPOP-Train generalize well to subpopulations that were not included during fine-tuning.

Consistent performance improvements over subpopulations. Figure 3 shows the per-group WD on the OpinionQA evaluation for Llama-2-7B, comparing our fine-tuning approach with zero-shot prompting and the empirical WD lower bound. To evaluate the consistency of performance gains, we calculate the *relative improvement* for each subpopulation as how much of the gap between zero-shot prompting and the empirical lower bound is reduced after fine-tuning. This measure allows us to account for varying lower bounds across subpopulations: since some groups have fewer respondents, there is greater uncertainty in their reported distribution in the survey data and greater variance between the original sample and bootstrap samples.

All 22 subpopulations demonstrate a large relative improvement after fine-tuning, ranging from 38%–54%. The average relative improvement is 46.7% with a standard deviation of 4.4%. This consistency confirms that our fine-tuning approach delivers balanced performance gains without disproportionately favoring any particular demographic subgroup. We hypothesize that the consistent gains over groups largely stem from our dataset design, which allocates an equal number of training samples to each group. By ensuring uniformly distributed data points across subpopulations, the model captures sufficient subgroup-specific signals, ultimately leading to consistent performance improvements.

Generalization to unseen subpopulations. We further investigate how models fine-tuned with our approach and SubPOP might show generalization to subpopulations that were not represented in the training data, a circumstance that may arise in real-world survey development. For the evaluation, we benchmark our methods against a zero-shot prompting baseline. Specifically, we evaluate our model, which is fine-tuned on 22 subpopulations provided in SubPOP-Train, on a set of subpopulations in OpinionQA that were not included in fine-tuning. This experiment not only checks generalization to unseen subpopulations, but also involves unseen survey questions, providing a robust assessment of the model capability for generalization to out-of-distribution data.

As shown in Table 2, our model achieves a strong reduction in WD even for unseen subpopulations, indicating that the model can be steered by demographic prompts beyond the seen subpopulations in training. Interestingly, although SubPOP-Train does not contain any data with opinion distributions of particular age groups (*e.g.* subjects of age 18-29 or those of age 65+), the average relative improvement is 44.7%, which is compatible with the average

Table 2: Per-group evaluation performance of Llama-2-7B-SubPOP-FT (Ours) on OpinionQA. We report the lower bound, WD for zero-shot prompting, WD for Llama-2-7B-SubPOP-FT, and the relative improvement. Rows highlighted in blue represent subpopulations included during fine-tuning, while uncolored rows correspond to subpopulations that were unseen during fine-tuning.

Group	Lower Bound	Zero Shot	Ours	Relative Improvement (%)
Age: 18-29	0.023	0.185	0.096	54.9
Age: 30-49	0.014	0.151	0.093	42.3
Age: 50-64	0.014	0.154	0.101	37.9
Age: 65+	0.013	0.195	0.115	44.0
Less than high school	0.043	0.161	0.101	50.8
High school graduate	0.017	0.144	0.092	40.9
Some college, no degree	0.018	0.144	0.093	40.5
Associate's degree	0.026	0.159	0.098	44.9
College grad	0.018	0.165	0.099	44.9
Postgraduate	0.015	0.174	0.106	42.8
Very conservative	0.026	0.208	0.107	55.5
Conservative	0.021	0.169	0.103	44.6
Moderate	0.016	0.151	0.094	42.2
Liberal	0.022	0.192	0.100	54.1
Very liberal	0.025	0.202	0.111	51.4
Democrat	0.016	0.172	0.099	47.1
Republican	0.019	0.196	0.105	52.0
Independent	0.016	0.155	0.093	44.5
Something Else	0.026	0.162	0.092	51.0

relative improvement for seen subpopulations. We provide results for other unseen groups in Table 7 of Appendix C.3 (average relative improvement of 43.1% with a standard deviation of 6.7%).

Steerability towards subpopulations. Given the large improvements in WD across subpopulations after fine-tuning, we want to test whether the LLM is truly adapting its predictions based on the subpopulation specified in its prompt (*i.e.* the LLM is being steered) or if the improvements can be explained by the LLMs' predictions getting closer to human responses in general, without any subpopulation-specific adaptation. If the LLM is being steered, we should expect that the LLM's predictions for a target subpopulation g_t are closer to the human distribution for g_t when g_t is the subpopulation specified in the prompt, compared to when another group g_s is specified in the prompt. We should also expect the gap in WD to be larger if the distance between the true human distributions for g_t and g_s are larger, such as differences between the youngest and oldest age groups compared to adjacent groups.

Formally, we define the *intergroup disagreement* between a target group g_t and a source group g_s as $\mathcal{WD}(p_H(\mathcal{A}_q \mid q, g_t), p_H(\mathcal{A}_q \mid q, g_s))$ averaged over evaluation questions. In human responses (left of Figure 4), the disagreement shows the pattern of locality: increases as the disparity in education levels between two groups grows. We extend this notion to compare the human distribution from the target group g_t with the LLM-predicted distribution when the *source* group g_s is specified in the prompt, $\mathcal{WD}(p_H(\mathcal{A}_q \mid q, g_t), p_\theta(\mathcal{A}_q \mid q, g_s))$. If the model truly incorporates subpopulation information from the prompt, its intergroup disagreement pattern should mirror that of the human data.

Zero-shot prompting with the base model (right of Figure 4) does not exhibit the locality pattern seen in the human data, indicating that it cannot be steered by subpopulation labels. In contrast, the fine-tuned model (middle of Figure 4) reproduces a pattern resembling the human-human case, even though it was trained on only two education groups ("less than high school" and "college graduate/some postgrad") and the other four groups were unseen. This result demonstrates that our fine-tuned model not only learns to condition on subpopulation information but also generalizes to subpopulations unseen during fine-tuning. We provide the intergroup disagreement for other traits in Appendix C.3.

4.4 Effect of Scaling the Dataset

In this section, we examine performance scales with training dataset size. We randomly sample subsets containing 25%, 50%, 75%, and 87.5% of the full SubPOP training set and evaluate three models—Llama-2-7B, Llama-2-13B, and Mistral-7B—on OpinionQA. As shown in Figure 5, we observe diminishing marginal returns, as is typical with fine-tuning; for example, after training on a random 25%, the models reach 72%-78% of the total improvement they achieve after fine-tuning on all of SubPOP-train. However, the performance does not entirely plateau. Instead, it continues to improve as we further increase the training data from 25% to 100%. We fit linear trend lines (dotted in Figure 5) to the results and observe that the slopes are similar for each model. This suggests that the rate of improvement—reflected by the slope in the power-law relationship—is intrinsic to the data and task rather than to the specific model architecture. In other words, LLMs exhibit comparable data efficiency, with performance gains that are fundamentally tied to dataset size rather than model-specific factors.

Using these trend lines, we can estimate the amount of fine-tuning data required to reach a target performance. For instance, we estimate that fine-tuning Mistral-7B on a dataset 25 times larger than the current SubPOP training set would yield a WD value of 0.07, which is much closer to the empirical lower bound of 0.031 reported in Table 1.

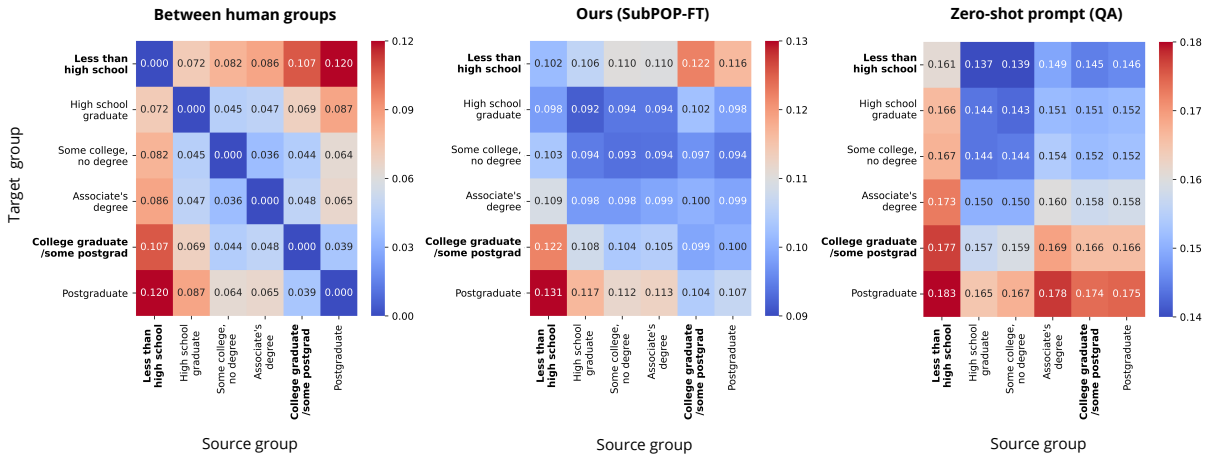

Figure 4: *Intergroup disagreement* pattern between groups of different education levels calculated with OpinionQA and Llama-2-7B as a base model. A target human group is compared to (left) a source human group, (middle) our fine-tuned model conditioned on a source group, (right) a base model conditioned on a source group. Bold-faced groups are included in the fine-tuning data SubPOP-Train, while the others aren't. In the human response (left), we observe a decreasing disagreement level as the education level becomes similar. This disagreement pattern exists in our fine-tuned model but not in the zero-shot prompting with a base model, indicating that our model can be steered to given subpopulation label even for unseen demographics while the base model cannot.

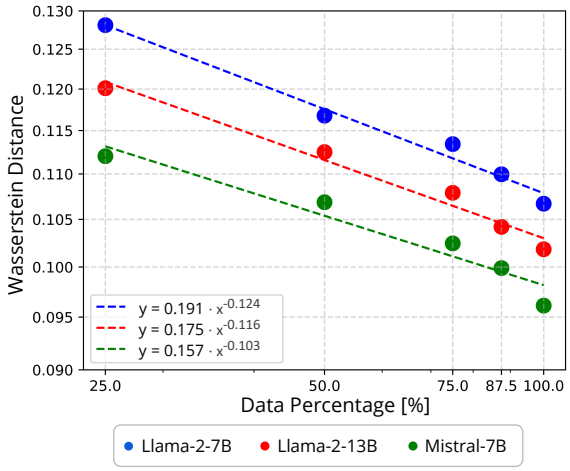

Figure 5: Evaluation results on OpinionQA after fine-tuning each LLM on increasingly large sampled subsets of SubPOP-Train. Both axes are presented in a log scale. The x-axis is the size of sampled dataset and the y-axis is WD against human responses measured on OpinionQA. Dashed lines represent a line of best fit. Performances at data percentage of 100% are identical to ours (SubPOP-FT) in Table 1.

This result underscores the critical importance of collecting more high-quality data, as increased dataset size can drive significant improvements in model performance.

5 Conclusion

In this work, we demonstrated that fine-tuning large language models on structured public opinion survey data markedly improves their ability to predict human response distributions. We curate SubPOP —a dataset 6.5× larger than previous collections to fine-tune and evaluate LLMs on survey response distribution prediction task. By fine-tuning on SubPOP, we showed that LLMs can capture the nuanced, group-specific variability in public opinions, while also generalizing to unseen subpopulations, survey waves and question topics, and different survey families. Fine-tuning achieves consistent improvements across subpopulations of varying sizes, and our experiments demonstrate that fine-tuned LLMs are indeed *adapting* their responses to the subpopulation specified in the prompt, even for subpopulations unseen during fine-tuning. Finally, our experiments also reveal that as the fine-tuning dataset grows, model performance continues to scale favorably, underscoring the value of our larger dataset.

Generalization is a critical capability for LLMs, if they are to be used to assist public opinion research, as researchers are most in need of accurate opinion predictions for questions or subpopulations whom they have not surveyed before. Our work, by greatly improving LLMs' ability to accurately predict opinions with fine-tuning and demonstrating strong generalization to out-of-distribution data, moves us closer towards the goal of leveraging LLMs for opinion prediction. However, many open questions remain: why is the model able to generalize well to unseen subpopulations and questions, and under what conditions might it fail to do so? How do we ensure that LLMs faithfully capture opinions along other dimensions that are not explored in this work, such as intersections of demographic identities or changing opinions over time? How should LLMs be integrated into survey designs, to serve as tools that can complement large-scale surveys with human participants? Answering these questions will require interdisciplinary collaborations with